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assignment 3

EXPLORER

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- cohere_example.py
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- groq_example.py
- huggingface_example.py
- ollama_example.py
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- requirements.txt
- SETUP.md

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cohere_example.py > ...

```python
def query_api(prompt):
    try:
        response = co.chat(
            model="command-xlarge-nightly",
            message=prompt,
            max_tokens=200,
            temperature=0.7
        )
        return response.text.strip()
    except Exception as e:
        return f"Error: {str(e)}"

if __name__ == "__main__":
    user_prompt = input("Enter prompt: ")
    print(query_api(user_prompt))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

powershell

```
API KEY: hEC40aNF6YkzfnngSuhTBvfZUGUPIAteZgWOGnHd
Enter prompt: what is ai?
AI, or Artificial Intelligence, refers to the simulation of human intelligence in machines that are programmed to think, learn, and make decisions like h
umans. These machines are designed to perform tasks that typically require human intelligence, such as understanding natural language, recognizing patter
ns, solving problems, and making decisions.

AI can be broadly categorized into two types:

1. **Narrow AI (Weak AI)**: This type of AI is designed to perform a specific task or a set of tasks. Examples include voice assistants like Siri or Alex
a, recommendation systems on platforms like Netflix or Amazon, and image recognition software. Narrow AI operates within a limited context and does not p
ossess general intelligence.

2. **General AI (Strong AI)**: This refers to machines that exhibit intelligence across a wide range of tasks, similar to human intelligence. General AI
can understand, learn, and apply knowledge in a way that is indistinguishable from a human being. As of now, General AI remains theoretical and has not b
een achieved.
PS C:\Users\tanc2\OneDrive\Desktop\A21\intership campusPe\assignment 3>
```

OUTLINE

TIMELINE

ZIP EXPLORER

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gemini_example.py > ...

```python
def query_api(prompt):
        response = model.generate_content(prompt)
        return response.text
    except Exception as e:
        return f"Error: {str(e)}"
if __name__ == "__main__":
    user_prompt = input("Enter prompt: ")
    print(query_api(user_prompt))
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

powershell

```
PS C:\Users\tanc2\OneDrive\Desktop\A21\intership campusPe\assignment 3> py gemini_example.py
C:\Users\tanc2\OneDrive\Desktop\A21\intership campusPe\assignment 3\gemini_example.py:2: FutureWarning:

All support for the `google.generativeai` package has ended. It will no longer be receiving
updates or bug fixes. Please switch to the `google.genai` package as soon as possible.
See README for more details:

https://github.com/google-gemini/deprecated-generative-ai-python/blob/main/README.md

  import google.generativeai as genai
Enter prompt: what is ai?
**Artificial Intelligence (AI)** is a broad field of computer science focused on creating machines that can perform tasks that traditionally require human intelligence.

In simpler terms, AI aims to make computers "think" and "learn" in ways similar to how humans do, enabling them to solve problems, understand language, recognize pattern
s, make decisions, and even create.

Here's a breakdown of what that entails:

1.  **Mimicking Human Cognition:** AI systems are designed to simulate various aspects of human intelligence, such as:
    *   **Learning:** Acquiring knowledge from data and experience without explicit programming.
    *   **Reasoning:** Using logic and rules to draw conclusions and make decisions.
    *   **Problem-Solving:** Finding solutions to complex challenges.
    *   **Perception:** Interpreting sensory information like images, sounds, and text.
    *   **Understanding Natural Language:** Communicating with humans in their own language.

2.  **Key Techniques and Subfields:**
    *   **Machine Learning (ML):** This is the most common way AI is implemented today. ML algorithms enable systems to learn from data, identify patterns, and make pred
```

OUTLINE
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ZIP EXPLORER

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groq_example.py ● | ollama_example.py | huggingface_example.py | gemini_example.py | cohere_example.py | README.md | SETUP.md | require

groq_example.py > ...

```python
import os
from groq import Groq
from dotenv import load_dotenv

load_dotenv()
api_key = os.getenv("GROQ_API_KEY")
if not api_key:
    print("API key not found!")
    exit()
client = Groq(api_key=api_key)

def query_api(prompt):
    try:
        response = client.chat.completions.create(
            model="llama-3.1-8b-instant",
            messages=[
                {"role": "user", "content": prompt}
            ],
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

powershell

```
PS C:\Users\tanc2\OneDrive\Desktop\A21\intership campusPe\assignment 3> py groq_example.py
Enter prompt: waht is ai?
AI, or Artificial Intelligence, refers to the simulation of human intelligence in machines that are programmed to think and learn like humans. The term may also be appli
ed to any machine that exhibits traits associated with a human mind such as learning and problem-solving.

AI technology is generally divided into two categories:

1. **Narrow or Weak AI**: This type of AI is designed to perform a specific task, such as facial recognition, language translation, or playing chess. Narrow AI systems a
re typically trained on large datasets and can improve their performance over time.

2. **General or Strong AI**: This type of AI is designed to perform any intellectual task that a human can. General AI systems are still in the early stages of developme
nt and are often referred to as "Artificial General Intelligence" or AGI.

Some common applications of AI include:

1. **Virtual Assistants**: AI-powered virtual assistants, such as Siri, Alexa, and Google Assistant, use natural language processing (NLP) to understand and respond
PS C:\Users\tanc2\OneDrive\Desktop\A21\intership campusPe\assignment 3>
```

OUTLINE
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⊗ 0 ⚠ 0 | Ln 5, Col 14 | Spaces: 4 | UTF-8 | CRLF | {} Python | 3.14.3 | Python 3.14.3 | Go Live

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huggingface_example.py > query_api

```python
from huggingface_hub import InferenceClient

load_dotenv()

api_key = os.getenv("HUGGINGFACE_API_KEY")

print("API KEY:", api_key)

if not api_key:
    print("API key not found!")
    exit()

client = InferenceClient(token=api_key)

def query_api(prompt):
    try:
        response = client.chat_completion(
            model="meta-llama/Meta-Llama-3-8B-Instruct",  # ✅ stable
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

powershell

```
PS C:\Users\tanc2\OneDrive\Desktop\A21\intership campusPe\assignment 3> py huggingface_example.py
API KEY: 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
Enter prompt: waht is ai?
AI, or Artificial Intelligence, refers to the simulation of human intelligence in machines that are programmed to think and learn like humans. The term can also be appli
ed to any machine that exhibits traits associated with a human mind such as learning and problem-solving.

AI technology is based on the principle of developing algorithms that can process data, identify patterns, and make decisions. These algorithms can be trained on large d
atasets to enable the AI system to learn from experience and improve its performance over time.

There are several types of AI, including:

1. **Narrow or Weak AI**: This type of AI is designed to perform a specific task, such as playing chess or recognizing faces. It is trained on a particular dataset and i
s not capable of general reasoning or problem-solving.
2. **General or Strong AI**: This type of AI is designed to perform any intellectual task that a human can. It is capable of reasoning, problem-solving, and learning in
the same way as humans.
3. **Superintelligence**: This
PS C:\Users\tanc2\OneDrive\Desktop\A21\intership campusPe\assignment 3>
```

OUTLINE

TIMELINE

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ollama_example.py > query_api

```python
def query_api(prompt):
    try:
        headers = {}
        if ollama_api_key:
            headers["Authorization"] = f"Bearer {ollama_api_key}"

        response = requests.post(
            "http://localhost:11434/api/generate",
            headers=headers,
            json={
                "model": "phi3",
                "prompt": prompt,
                "stream": False
            },
            timeout=30
        )

        if response.status_code >= 400:
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\tanc2\OneDrive\Desktop\A21\intership campusPe\assignment 3> py ollama_example.py
Enter prompt: what is ai?
AI, or artificial intelligence, refers to the simulation of human-like intelligence in machines that are programmed to think and act like humans. It encompdeces a wide range o
f technologies designed for tasks such as visual perception, speech recognition, decision making, problem solving, and language translation among others. AI systems can be cat
egorized into two types: narrow or weak AI, which is created with specific purposes in mind (like voice assistants), and general or strong AI that has the ability to understan
d, learn, adapt, plan, solve problems, and unprompted carry out tasks requiring human intelligence.
PS C:\Users\tanc2\OneDrive\Desktop\A21\intership campusPe\assignment 3>
```

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OUTLINE
TIMELINE
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